
Does AI-Assisted Writing Resurrect Retracted Science?

A Population-Level Study of Zombie Citations

Anonymous Author(s)

Affiliation

Address

email

Abstract

Large language models trained before a retraction cannot know about it, and chatbot-level audits have shown that conversational models draw on retracted articles in a majority of relevant answers while flagging the retraction less than half the time. Whether this failure reaches the published literature is unknown. We measure it, combining the Retraction Watch database with a frozen OpenAlex snapshot to track citations to already-retracted papers, which we call zombie citations, across 204.9 million citing works from 2018 to 2026. Our design has three prongs, each informative on its own: an interrupted time series of the population zombie-citation rate around the release of ChatGPT, a matched-cohort comparison of 1,731 papers containing verbatim LLM artifacts against controls matched on venue, year, and citedness, and partial-identification bounds that convert a noisy lexical measure of AI involvement into an honest interval rather than an attenuated point estimate. The three prongs disagree in a way that is itself the finding. The population rate rises by 0.232 zombie citations per 10,000 references at the ChatGPT release, a 20% increase over the pre-period rate of 1.160, exceeding all 36 placebo breaks and concentrated in computational fields. Yet papers whose LLM involvement is certain cite retracted work no more often than their matched controls, at a ratio of 0.878 with a 95% interval of 0.379 to 1.632, while the lexical proxy appears to separate sharply until reference count is held fixed. Flagged papers carry 44.7 references on average against 9.2 for the rest, and the outcome is a per-paper indicator, so most of the apparent contrast is article type. Comparing full articles to full articles leaves a modest elevation that reverses in the largest-bibliography stratum. No per-paper design we can build attributes the population shift to AI assistance, and the proxies that would claim otherwise measure who writes long articles. Data, code, and the artifact corpus are available at <https://github.com/garyzhang1006/ai-science-zombie-data>.

1 Introduction

Retraction is the scientific record’s error-correction mechanism, and it works only if authors check the current status of what they cite. Large language models complicate that mechanism in a specific, foreseeable way: a model whose training data was collected before a retraction has no representation of it, and a model asked to summarize a field will reproduce the retracted claim with the same fluency as any other. The chatbot-level evidence is already troubling, since a recent audit found that ChatGPT drew on 61.5% of retracted stem-cell articles when answering domain questions and mentioned the retraction in only 41.7% of those responses (Zhang et al., 2025), and a broader study reached similar conclusions across fields (Thelwall et al., 2025). What no study has measured is

36 whether the published literature itself now cites retracted work at an elevated rate because of AI-
37 assisted writing. That is the question this paper answers.

38 The measurement problem is harder than it looks, and the difficulty shapes our design. Citations
39 to retracted papers are rare, AI involvement is observed only through noisy proxies, and a naive
40 regression of zombie-citation rates on a lexical AI score is attenuated by proxy error, so a null from
41 that regression would describe the proxy rather than the literature. We therefore built the study
42 around three prongs whose failure modes do not overlap: a population interrupted time series that
43 needs no per-paper attribution at all; a cohort in which AI involvement is certain, because the papers
44 carry verbatim LLM artifacts such as “as an AI language model” that reach print only through
45 unedited pasting; and set-identified bounds that make the lexical score of Kobak et al. (2024) and
46 Liang et al. (2025) honest instead of discarding it, in the tradition of partial identification under
47 misclassification (Manski, 2003).

48 Running all three was worth the cost, because they disagree. The population series moves at the
49 release of ChatGPT. The cohort with certain exposure does not move at all. The lexical proxy
50 separates sharply until reference count is held fixed, and then barely. A study built on any one prong
51 would have reported a clean result and been wrong about what it measured.

52 In the same cohorts we track a second, denser outcome: references that do not resolve, which
53 prior evidence links to model fabrication far above the human baseline (Walters and Wilder, 2023;
54 Bhattacharyya et al., 2023). A zombie citation is real-but-refuted knowledge propagating; an unre-
55 solvable reference is fabricated support.

56 Our contributions are: (1) the first population-level measurement of zombie-citation dynamics across
57 the LLM transition, covering 204.9 million citing works and 216,514 zombie citation events; (2) the
58 artifact cohort, a public corpus of 1,731 papers with certain LLM involvement and matched controls,
59 together with a measured false-positive rate for every artifact phrase we use; (3) bounded rather than
60 attenuated estimates of the dose-response between lexical AI involvement and citation integrity,
61 computed within strata of reference-list length; and (4) a demonstration that the unconditioned ver-
62 sion of that estimate overstates the contrast roughly hundredfold, because excess-vocabulary proxies
63 select long articles, which is a warning about a measurement now in wide use.

64 2 Related work

65 Post-retraction citation is a longstanding integrity concern predating language models, covering the
66 persistence of citations to retracted work and interventions such as the RISRS recommendations
67 (Schneider et al., 2022). Case studies have traced the citation network of a single prominent retracted
68 paper and found direct citation to be the main propagation channel (van der Vet and Nijveen, 2016).
69 We add scale and a treatment contrast, measuring the phenomenon across the full indexed literature
70 and asking whether the LLM transition changed its rate.

71 On the AI side, audits of conversational models document both the retraction blindness described
72 above (Zhang et al., 2025; Thelwall et al., 2025) and the fabrication of bibliographic references,
73 which Walters and Wilder measured at 55% of GPT-3.5 citations and 18% of GPT-4 citations, with
74 substantive errors in a further 24% of the GPT-4 citations that do refer to real work (Walters and
75 Wilder, 2023; Bhattacharyya et al., 2023). Verbatim artifact phrases in published papers were cat-
76 alogued by Strzelecki across premier journals (Strzelecki, 2025), establishing that unedited model
77 output does reach print; we industrialize that observation into a matched-cohort design. Population-
78 level prevalence estimation (Kobak et al., 2024; Liang et al., 2025; Thelwall, 2025) provides our
79 dose proxy, and its error rates feed the bounds. To our knowledge no prior work connects these
80 threads to measured citation-integrity outcomes in the literature itself.

81 3 Data and design

82 3.1 Retraction and citation data

83 Retraction events come from the Retraction Watch database, which yields 60,247 retractions carry-
84 ing both a usable DOI and a retraction date, of which 59,924 (99.5%) match a work in OpenAlex.
85 Citation events and reference lists come from the OpenAlex parquet snapshot of 26 June 2026,
86 pinned to that single release so that every number here is reproducible against a frozen corpus rather

Table 1: Artifact phrase registry with false-positive rates measured against the pre-ChatGPT era. A phrase appearing in a paper published before 2023 cannot be unedited model output, so the pre-2023 share of hits estimates the phrase’s false-positive rate.

Phrase	Hits	Pre-2023	FP rate	Status
as an AI language model	9,131	111	1.2%	retained
as a large language model, I	244	1	0.4%	retained
as of my last knowledge update	157	3	1.9%	retained
as an AI developed by OpenAI	99	3	3.0%	retained
my knowledge cutoff	89	0	0.0%	retained
I don’t have access to real-time	22	1	4.5%	retained
I cannot browse the internet	3	0	0.0%	retained
Regenerate response	9,085	7,926	87.2%	excluded

than a moving index. We work from the snapshot rather than the REST API by necessity: the API now meters usage at 1,000 calls or \$0.10 per day on its free tier, which cannot cover a scan of half a billion works, and the snapshot is unmetered.

A citation is a zombie citation if the citing paper appeared at least six months after the cited paper’s retraction date, the buffer absorbing indexing lag, and we exclude citing works that mention retraction in their title or abstract, since citing a retracted paper to discuss its retraction is the correction system working as intended. Scanning all 2,446 snapshot files yields 216,514 zombie citation events across 204.9 million citing works published between 2018 and 2026. Both the numerator and the denominator come from the same pass, so the monthly rate is exact rather than estimated from a sample.

3.2 The artifact cohort

Candidate artifact papers are harvested by full-text search over a registry of verbatim LLM phrases. This step needs more care than it first appears, and getting it wrong would have destroyed the cohort. OpenAlex full-text search is stemmed, so the phrase “regenerate response” matches “regenerative response” and returns 9,085 hits dominated by zebrafish and stem-cell regeneration papers. Those hits would have made up 82% of our cohort.

We caught this with a negative control that the study period supplies for free. No paper published before 2023 can contain unedited ChatGPT output, so every pre-2023 hit is a false positive by construction, and the pre-2023 share of a phrase’s hits estimates its false-positive rate directly. Table 1 reports that rate for every phrase we screened. “Regenerate response” fails at 87.2% and is excluded. The seven surviving phrases range from 0.0% to 4.5%, with the workhorse phrase “as an AI language model” at 1.2% across 9,131 hits. This converts the claim that these phrases make model involvement certain from an assertion into a measurement.

Filtering then removes papers whose own subject is language models, since those quote the phrases deliberately: 3,613 candidates are dropped on title, 1,868 on subfield, and 1,549 for carrying no references. What remains is 1,731 papers, all published in 2023 or later, of which 1,643 carry a venue identifier. Each is matched to five controls from the same venue and year, within 25% of its reference-list length and nearest in citation count, giving 7,644 matched pairs. Reference-list lengths come out nearly identical across arms, 42.55 against 42.45, so the matching binds. Within this cohort the exposure is observed rather than proxied, at the price of representing only the least careful tail of AI-assisted writing, a selection we return to in Section 6.

3.3 Outcomes, series, and bounds

For every paper we compute two outcomes over its reference list: the count of zombie citations as defined above, and the count of unresolvable references, meaning entries whose DOI fails to resolve in Crossref and whose bibliographic string matches no indexed work under fuzzy search at a similarity threshold of 85. The population series aggregates the first outcome monthly within field strata, and the interrupted time series allows a level and slope change at the ChatGPT release

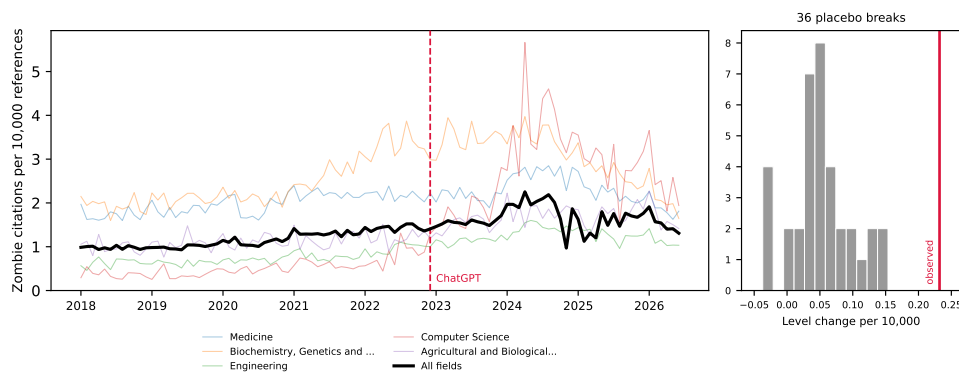


Figure 1: Zombie citations per 10,000 references, monthly, 2018–2026, for the five fields contributing the most events and for the literature as a whole. The vertical line marks the ChatGPT release; the inset shows the distribution of 36 placebo break estimates from 2019–2021 against the estimated 2022 break.

124 in December 2022, with Newey-West standard errors and placebo breaks at each month of 2019
125 through 2021 calibrating the procedure’s own false-positive rate.

126 The bounding analysis follows Manski (2003). Given a lexical AI score with assumed sensitivity
127 and specificity, the observed outcome-by-score table set-identifies the true outcome rates among AI-
128 assisted and unassisted papers. We solve this over a grid rather than at any single assumed point, and
129 the grid is disciplined by the data: our rule flags 0.86% of 2023-onward papers, and the prevalence
130 solution requires specificity above $1 - P(S=1) = 0.991$, so any specificity below that admits more
131 false positives than observed positives and no solution at all. A rule demanding two distinct markers
132 from a 37-term list is high-specificity and low-sensitivity by construction, and the grid sits in that
133 regime.

134 The outcome for this prong is a per-paper indicator, whether a paper carries at least one zombie
135 citation, and that indicator rises with the length of a paper’s reference list. The lexical rule needs two
136 markers inside an abstract, so it selects papers with long abstracts, which are full research articles
137 carrying many references. Across the unrestricted sample, flagged papers average 44.7 references
138 against 9.2 for the rest. We therefore compute the bounds within reference-count strata, and take
139 papers with at least 20 references as the primary specification, which compares full articles to full
140 articles at 61.2 references against 50.8.

141 4 Results

142 4.1 Population dynamics

143 The population rate of zombie citation runs at 1.160 per 10,000 references over the pre-period and
144 rises through it, and the interrupted time series estimates a level change of +0.232 at the ChatGPT
145 release (standard error 0.129, $p = 0.074$), a 20% increase on the pre-period rate. The conventional
146 p -value is marginal, but it is the weaker of the two tests available here. Across 36 placebo breaks
147 placed through 2019 to 2021, the largest absolute level change is 0.154, so the real break exceeds
148 every placebo the procedure produces on data where nothing happened, a placebo-based p below
149 $1/37$ (Figure 1).

150 The level change is stable under truncation: refitting with the series ending in June 2025, December
151 2025, March 2026, and June 2026 gives +0.193, +0.190, +0.194, and +0.232. The slope change
152 is not. It reads -0.0116 ($p = 0.005$) on the full series but decays to -0.0086 ($p = 0.312$) when the
153 last year is dropped, and the snapshot’s final months are visibly under-indexed, carrying 9.0 million
154 references in June 2026 against a monthly norm near 15 million. We therefore report the level shift
155 as the population result and treat the post-break decline as an artifact of the indexing tail.

156 The break concentrates where one would expect AI writing tools to have landed first. Computer
157 Science shows a level change of +1.489 ($p = 0.006$), against +0.606 in Biochemistry ($p = 0.043$),

Table 2: Artifact cohort against matched controls. Rates per 1,000 references; ratios with 95% cluster-bootstrap intervals resampling papers. The unresolvable-reference row rests on 7,873 papers and 415,750 references checked against Crossref.

Outcome	Artifact papers	Matched controls	Ratio
Zombie citations	0.157	0.179	0.878 [0.379, 1.632]
Unresolvable references	138.84	127.74	1.087 [1.005, 1.174]
Reference-list length	42.55	42.45	

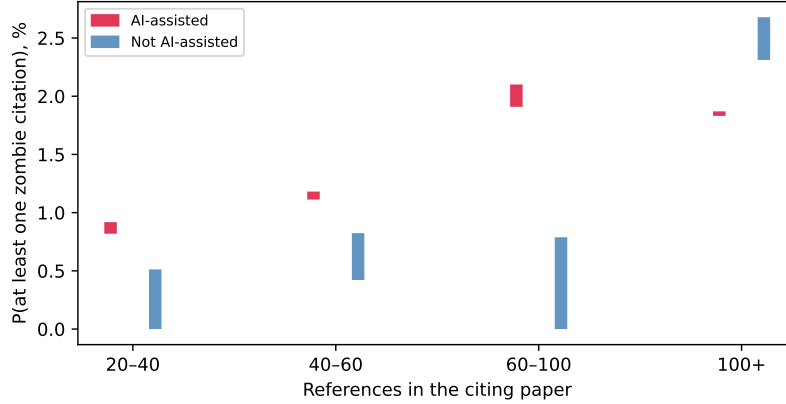


Figure 2: Set-identified probability of carrying at least one zombie citation, by lexical AI score, within strata of reference-list length. Bars span the full range of solutions across the sensitivity and specificity grid. The contrast is modest where it appears and reverses in the largest-bibliography stratum.

158 +0.296 in Engineering ($p = 0.010$), +0.297 in Agricultural and Biological Sciences ($p = 0.005$),
159 and +0.174 in Materials Science ($p = 0.017$). Medicine, the largest field by volume, does not move
160 (+0.163, $p = 0.362$). Eight fields were tested, and Computer Science ($p = 0.0057$) and Agricultural
161 and Biological Sciences ($p = 0.0045$) are the two that survive a Bonferroni threshold of 0.00625;
162 we read the rest as suggestive of a computational concentration rather than as separately established.

163 4.2 The artifact cohort

164 Within the cohort where LLM involvement is certain, the zombie-citation result is a null. Artifact
165 papers cite retracted work at 0.157 per 1,000 references against 0.179 among controls, a ratio of
166 0.878 whose 95% interval, 0.379 to 1.632, comfortably contains parity (Table 2). This is a bounded
167 null rather than an absence of evidence: the design rules out effects above roughly 1.6 times the
168 control rate, and the point estimate sits below one. The endpoint is simply sparse. At 0.157 per
169 1,000 references, 1,645 artifact papers carrying 42.55 references each are expected to produce about
170 eleven zombie citations in total, and no cohort of this size can resolve a modest effect on such a rate.

171 The denser endpoint carries what statistical weight the cohort has, as the design anticipated. Arti-
172 fact papers carry 138.84 unresolvable references per 1,000 against 127.74 among controls, a ratio of
173 1.087 with an interval of 1.005 to 1.174. The effect is real but small, and its interval touches parity
174 closely enough that we decline to describe it as robust. Its absolute level also deserves care: an unre-
175 solvable rate near 13% across both arms is far too high to represent fabrication, and mostly reflects
176 Crossref indexing gaps for books, theses, and items without DOIs. Only the ratio between matched
177 arms is interpretable, and only under the assumption that those indexing gaps fall symmetrically
178 across artifact and control papers drawn from the same venue and year.

4.3 Bounded dose-response

Applied without restriction to a seeded sample of 629,900 papers published from 2023 onward, this prong produces the largest contrast in the study and a misleading one. The lexical rule flags 0.86% of papers. Among those, 1.057% carry at least one zombie citation against 0.113% among the rest, and the correction set-identifies the flagged rate within [1.24%, 5.11%] against [0.00%, 0.089%], intervals that do not overlap anywhere on the grid. That comparison is mostly about article type. Flagged papers average 44.7 references against 9.2, because a rule requiring two markers inside an abstract selects long-abstract full articles, and the outcome counts whether a paper carries any zombie citation at all. A paper with sixty references has more opportunities to carry one than a paper with six, whoever wrote it.

Restricting to papers with at least 20 references brings the arms to 61.2 against 50.8 and shrinks the effect substantially. The naive contrast falls to 1.261% against 0.816%, a ratio of 1.55 rather than roughly nine, and the bounds give [1.27%, 1.40%] for flagged papers against [0.00%, 0.745%] for the rest. The correction now moves the estimate by a factor of 1.11 rather than the 13 the unrestricted fit suggested, so that attenuation was itself an artifact of composition.

Within strata the picture is neither large nor monotone (Figure 2). Flagged papers sit higher in three of four strata, at [0.82%, 0.92%] against [0.00%, 0.51%] for 20 to 40 references, [1.11%, 1.18%] against [0.42%, 0.82%] for 40 to 60, and [1.91%, 2.10%] against [0.00%, 0.79%] for 60 to 100. Above 100 references the ordering reverses, [1.83%, 1.87%] against [2.31%, 2.68%]. A dose-response that shrinks and then inverts as bibliographies grow is not the signature of a causal effect on citation checking.

The gap between the two fits is the transferable lesson. Excess-vocabulary proxies are increasingly used to correlate LLM prevalence with quality outcomes, and any such correlation inherits whatever the markers track. Here they track article length strongly enough to manufacture a hundredfold contrast out of a modest one, and misclassification bounds do not help: the correction assumes the proxy errs at random with respect to the outcome, and length-driven selection violates that in the direction that inflates the result.

5 Discussion

The three prongs give three different answers, and the disagreement is the paper’s substance. The population rate moves at the ChatGPT release by an amount that survives every placebo the data can generate. Papers whose LLM involvement is certain show nothing on the retracted-citation endpoint and a small effect on the denser one. The lexical proxy appears to separate sharply and stops doing so once reference count is held fixed.

The reconciliation we favor is that no per-paper design available to us attributes the population shift to AI assistance, and that the design which looked like it did was measuring article length. The cohort where exposure is observed rather than inferred is where a causal effect should be largest, and it is absent there. What the proxy separates is a population that differs from the rest of the literature in more ways than its use of language models, most obviously in how much it cites.

It is worth asking whether the proxy route could at least account for the population shift arithmetically, and the answer shows why it cannot settle anything. Among papers with at least 20 references, flagged papers carry zombie citations at 2.31 per 10,000 references against 1.73 for the rest. Scaling that excess by the AI prevalence each assumed sensitivity implies gives an attributable shift of 0.097 per 10,000 at a sensitivity of 0.20, 0.193 at 0.10, and 0.387 at 0.05, which is 42%, 83%, and 167% of the observed break. The same data explains half the break or more than all of it depending on a parameter no one has measured, so this route cannot discriminate even before the confounding is considered.

This leaves the population result standing and unexplained, which we think is the honest place to leave it. The break is concentrated in computational fields, survives 36 placebo breaks and four series endpoints, and coincides with the arrival of AI writing tools. It is equally consistent with contemporaneous changes in indexing, in citation practice, or in who was publishing in those fields in 2023. Distinguishing these needs per-paper exposure that is neither self-reported nor lexical, which is a real gap we cannot close with public data.

231 The artifact-cohort null needs one qualification. The cohort is selected on carelessness, since a
232 paper that ships “as an AI language model” verbatim received no meaningful proofreading, so its
233 null bounds the harm of the careless tail rather than of AI assistance in general. That tail is also
234 where integrity screening would go first, which makes the null directly useful to anyone planning
235 such screening.

236 For the workshop’s framing question, the study offers a measured instance of a general worry and a
237 sharper caution about how the worry gets measured. The aggregate rate at which the literature cites
238 retracted work did shift when AI writing tools arrived. Whether AI assistance caused it remains
239 unresolved by our strongest design, and the lexical proxies now in wide circulation would have
240 answered the question confidently and, on this evidence, wrongly.

241 6 Limitations

242 The unresolvable-reference comparison queried every one of the 9,009 intended papers, of which
243 7,873 (87.3%) deposit reference data in Crossref at all; the remainder are absent from the com-
244 parison for a reason unrelated to their content. OpenAlex full-text coverage is partial and skewed
245 toward open access, so the artifact cohort undercounts paywalled carelessness. The artifact filter’s
246 precision is established by the era-based false-positive rates in Table 1 rather than by manual adju-
247 dication, which we did not perform; the era test bounds how often a phrase fires on text no model
248 could have written, and says nothing about papers where the phrase is quoted by a 2023-or-later
249 author discussing model output, which the topic and title filters target instead. We publish the full
250 candidate list with every drop reason, together with a seeded 100-paper sample, so that any reader
251 can audit both. Retraction Watch coverage is itself uneven across fields and eras. The interrupted
252 time series identifies a break at the ChatGPT release but cannot by itself attribute the break to LLM
253 writing rather than to contemporaneous changes in indexing or citation practice, which the placebo
254 distribution mitigates without eliminating. The bounds correct for misclassification of the AI proxy
255 and not for confounding, and Section 4 shows that the confounding is large enough to dominate the
256 unconditioned estimate; the conditioned estimate holds reference count fixed but nothing else, so
257 field, venue, and author language remain uncontrolled. Finally, all three prongs measure citation to
258 retracted work, and none measures whether the citing text endorses the retracted claim, so our rates
259 are upper bounds on endorsement.

260 7 Conclusion

261 The literature’s zombie-citation rate rose about 20% when AI writing tools arrived, against a pre-
262 period rate of 1.160 per 10,000 references, and the papers we can prove were written with a language
263 model show no such elevation. We cannot close that gap with public data, and we think saying so is
264 more useful than picking whichever prong tells a cleaner story. The measurement lesson generalizes
265 past this question: an excess-vocabulary proxy separated the literature by a factor of roughly a
266 hundred on citation integrity, and nearly all of it was article length. The artifact cohort with its
267 measured per-phrase false-positive rates, the bounds machinery, and the monthly series are released
268 at <https://github.com/garyzhang1006/ai-science-zombie-data>.

269 References

- 270 X. Zhang, T. Gu, S. Loganathan, Y. Xie, and J. Chen. Retracted articles on stem cells are contin-
271 uously cited in publications and used by ChatGPT. *Scientometrics*, 130(11):6503–6511, 2025.
272 doi:10.1007/s11192-025-05484-y
- 273 M. Thelwall, M. Lehtisaari, I. Katsirea, K. Holmberg, and E. Zheng. Does ChatGPT ig-
274 nore article retractions and other reliability concerns? *Learned Publishing*, 38(4), 2025.
275 doi:10.1002/leap.2018
- 276 A. Strzelecki. ‘As of my last knowledge update’: How is content generated by ChatGPT in-
277 filtrating scientific papers published in premier journals? *Learned Publishing*, 38(1), 2025.
278 doi:10.1002/leap.1650
- 279 D. Kobak, R. González-Márquez, E. Á. Horvát, and J. Lause. Delving into LLM-assisted writing in
280 biomedical publications through excess vocabulary. *arXiv:2406.07016*, 2024.

- 281 W. Liang, Y. Zhang, Z. Wu, et al. Quantifying large language model usage in scientific papers.
282 *Nature Human Behaviour*, 9(12):2599–2609, 2025. doi:10.1038/s41562-025-02273-8
- 283 M. Thelwall. ChatGPT contamination: estimating the prevalence of LLMs in the scholarly literature.
284 *arXiv:2403.16887*, 2025.
- 285 W. H. Walters and E. I. Wilder. Fabrication and errors in the bibliographic citations generated by
286 ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi:10.1038/s41598-023-41032-5
- 287 M. Bhattacharyya, V. M. Miller, D. Bhattacharyya, and L. E. Miller. High rates of fab-
288 ricated and inaccurate references in ChatGPT-generated medical content. *Cureus*, 2023.
289 doi:10.7759/cureus.39238
- 290 C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.
- 291 J. Schneider, N. Woods, R. Proescholdt, et al. Reducing the inadvertent spread of retracted science:
292 recommendations from the RISRS report. *Research Integrity and Peer Review*, 7(1):6, 2022.
293 doi:10.1186/s41073-022-00125-x
- 294 P. E. van der Vet and H. Nijveen. Propagation of errors in citation networks: a study involving the
295 entire citation network of a widely cited paper published in, and later retracted from, the journal
296 *Nature*. *Research Integrity and Peer Review*, 1(1):3, 2016. doi:10.1186/s41073-016-0008-5

297 **A Artifact phrase registry and filtering**

298 Table 1 gives the full registry with raw counts and measured false-positive rates. The filter drops can-
299 didates in this order, recording a reason for each: phrase retired for stem collision (9,088), published
300 before 2023 (119), subject classified in an AI subfield (1,868), title matching a language-model topic
301 pattern (3,613), abstract carrying two or more meta-discussion terms (125 across variants), document
302 type outside the article, review, chapter, preprint, and dissertation set (approximately 500), and no
303 reference list (1,549). The complete candidate table with per-paper decisions ships in the reposi-
304 tory as `artifact_candidates.csv`, and a seeded 100-paper sample for hand validation ships as
305 `hand_validation_sample.csv`.

306 **B Robustness**

307 The series is refit at four endpoints in Section 4; the level change varies between +0.190 and +0.232
308 while the slope change loses significance, which is why we report only the level shift. The zombie
309 definition uses a six-month buffer after the retraction date; the buffer, the matching tolerance of
310 25%, and the fuzzy-match threshold of 85 are set in `config.py` and can be varied by rerunning
311 the pipeline. The misclassification grid spans sensitivities of 0.03 to 0.20 and specificities of 0.992
312 to 0.999, and all sixteen combinations admit a solution; the identification constraint that fixes the
313 lower edge of the specificity range is derived in Section 3. Bounds use reference-count strata, with
314 at least 20 references as the primary cut; `bounds.json` reports the unrestricted fit, the primary fit,
315 and every stratum.